

Transformations with Multiple Parameters

Answer Key

Precalculus Functions and Composition

Quick Answers

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|--------------|------------------|
| 1. 23 | 6. 4 |
| 2. -4 | 7. $-4 \cos(3x)$ |
| 3. 21 | 8. -1 |
| 4. 5 | 9. 3 |
| 5. $3x + 15$ | 10. — |

Common wrong answers

- If they got 143: moved the input the way the printed sign reads, using $x + 2$ in place of $x - 2$*
- If they got 14: applied the reflection to the input before squaring instead of to the output after it*
- If they got 4.12: substituted without carrying the horizontal compression through the parentheses, evaluating the sine at x minus the phase shift instead of at twice that difference*
- If they got 3: left the horizontal factor out of the input, taking the root of x minus 1 instead of 4 times it*

Throughout, parameters are read against $g(x) = a f(b(x - h)) + k$: a and k act on outputs and do what they say; b and h act on inputs and do the opposite, and only after the inside is factored into $b(x - h)$. The “common wrong answers” block above names the parameter each wrong value mishandled.

1. $f(x) = x^2$, $g(x) = 3f(x - 2) - 4$; **complete the chain and find $g(5)$.**

The chain is the parameter order made visible — input shift, then the parent, then the output stretch, then the output shift:

x	$x - 2$	$f(x - 2)$	$3f(x - 2)$	$3f(x - 2) - 4$
5	3	9	27	23

The -2 is applied to the *input* before f ever runs, which is why the table’s second column turns 5 into 3. $g(5) = 23$

If they got 143: they moved the input the way the sign reads and used $x + 2$.

2. $h(x) = -f(x + 2) + 5$ with $f(x) = x^2$; name the parameters, find $h(1)$.

Two parameters are non-trivial: $a = -1$ (a reflection across the x -axis, an *output* move) and $h = -2$ (a shift 2 units *left*, an input move — the inside reads $x + 2 = x - (-2)$). There is no stretch ($b = 1$) and the vertical shift is $k = 5$.

$$h(1) = -(1 + 2)^2 + 5 = -9 + 5$$

$$\boxed{h(1) = -4}$$

If they got 14: they reflected the input before squaring, computing $-(1 + 2)^2 + 5$. Squaring happens first; the minus sign belongs to the output.

3. Solve $2\sqrt{x - 5} + 3 = 11$.

Undo the parameters in the reverse of the order that builds them — outputs first, inputs last:

$$\begin{array}{ll} 2\sqrt{x - 5} + 3 = 11 & \text{undo } k = 3 : \\ 2\sqrt{x - 5} = 8 & \text{undo } a = 2 : \\ \sqrt{x - 5} = 4 & \text{undo the parent (square both sides):} \\ x - 5 = 16 & \text{undo } h = 5 : \\ x = 21 & \end{array}$$

Check: $2\sqrt{21 - 5} + 3 = 2(4) + 3 = 11$ ✓

$$\boxed{x = 21}$$

4. $y = 3\sin(2(x - \pi/4)) + 2$: parameters, then y at $x = \pi/2$.

Amplitude = $|a| = 3$. Period = $2\pi/b = 2\pi/2 = \pi$. Phase shift = $h = \pi/4$ to the right. Midline = $y = k = 2$.

The inside is already factored, so substituting is safe — but the factor $b = 2$ has to travel through the parentheses:

$$\begin{aligned} 2\left(\frac{\pi}{2} - \frac{\pi}{4}\right) &= 2 \cdot \frac{\pi}{4} = \frac{\pi}{2} \\ y &= 3\sin\frac{\pi}{2} + 2 = 3(1) + 2 \end{aligned}$$

That input is exactly a quarter period past the phase shift, which is where a sine reaches its maximum — so 5 should be the largest value on the graph, and it is. $\boxed{y = 5}$

Common wrong answer ≈ 4.12 : the student dropped the factor 2 and evaluated $3\sin(\pi/2 - \pi/4) + 2$, which is a real number on no graph in this problem.

5. $s(x) = x + 5$, $d(x) = 3x$; simplify $(d \circ s)(x)$.

$(d \circ s)(x) = d(s(x))$ means s runs *first*: the shift happens, then the stretch acts on the already-shifted value.

$$d(s(x)) = d(x + 5) = 3(x + 5)$$

$$\boxed{(d \circ s)(x) = 3x + 15}$$

Notice the shift got multiplied by the stretch factor: shifting 5 and then tripling moves the graph 15, not 5. That is the whole reason order matters.

6. Solve $(d \circ s)(x) = 27$, and why $(s \circ d)$ differs.

From problem 5, $(d \circ s)(x) = 3(x + 5)$:

$$3(x + 5) = 27 \Rightarrow x + 5 = 9 \Rightarrow x = 4$$

$$\boxed{x = 4}$$

Why the other order differs: $(s \circ d)(x) = s(3x) = 3x + 5$. Here the stretch happens first, so the shift is applied afterwards and is *not* scaled. Setting $3x + 5 = 27$ would give a different x entirely. Same two parameters, different composition order, different function.

7. Rewrite $y = 4 \sin(3x - \pi/2)$ as $A \sin(B(x - C))$.

C cannot be read off until B is factored out of the whole inside expression:

$$3x - \frac{\pi}{2} = 3 \left(x - \frac{\pi}{6} \right)$$

because $3 \cdot \pi/6 = \pi/2$. So

$$\boxed{y = 4 \sin \left(3 \left(x - \frac{\pi}{6} \right) \right)}, \quad A = 4, \quad B = 3, \quad C = \frac{\pi}{6}$$

Why $C \neq \pi/2$: the $\pi/2$ is subtracted *after* the input has already been tripled, so it measures a shift in the compressed variable, not in x . Dividing by B converts it back: $C = (\pi/2)/3 = \pi/6$.

8. $g(x) = -2f(4(x - 1)) + 7$ with $f(x) = \sqrt{x}$; parameters and $g(5)$.

Four parameters: $a = -2$ (reflect across the x -axis and stretch outputs by 2), $b = 4$ (compress inputs horizontally by a factor of $1/4$), $h = 1$ (shift 1 unit right), $k = 7$ (shift 7 units up).

$$g(5) = -2\sqrt{4(5 - 1)} + 7 = -2\sqrt{16} + 7 = -2(4) + 7 = -8 + 7$$

$$\boxed{g(5) = -1}$$

Order check: the input move $4(5 - 1) = 16$ happens before the square root; the -2 and the $+7$ act only on the output the parent returned.

If they got 3: they left b out of the input and took $\sqrt{5 - 1}$ instead.

9. x -intercept of $g(x) = 3\sqrt{x + 6} - 9$.

Set the output to 0 and peel the parameters off from the outside in:

$$\begin{array}{ll} 3\sqrt{x + 6} - 9 = 0 & \\ 3\sqrt{x + 6} = 9 & \text{undo } k = -9 \\ \sqrt{x + 6} = 3 & \text{undo } a = 3 \\ x + 6 = 9 & \text{square} \\ x = 3 & \text{undo } h = -6 \end{array}$$

$$\text{Check: } 3\sqrt{3 + 6} - 9 = 3(3) - 9 = 0 \checkmark$$

$$\boxed{x = 3}$$

10. Sketch $y = -3f\left(\frac{x}{3} + 1\right) + 2$ **for** $f(x) = \sqrt{x}$. (open construction — verify the sketch and read the ordering argument)

Step 1 — factor the inside. $\frac{x}{3} + 1 = \frac{1}{3}(x + 3)$, so $b = \frac{1}{3}$ and $h = -3$.

Step 2 — the four parameters, in a correct order of application: horizontal stretch by 3 (since $1/b = 3$), then shift 3 units left, then vertical stretch by 3 with a reflection across the x -axis, then shift 2 units up. The two input moves must be done before the output moves, and within the input moves the stretch is applied before the shift when the inside is written in factored form.

Step 3 — track the parent's starting point. f starts at $(0, 0)$. The inside is zero when $x = -3$, and the output there is $-3(0) + 2 = 2$, so the image of the starting point is $(-3, 2)$ — the sketch must begin there.

Step 4 — shape. From $(-3, 2)$ the curve falls to the right (because $a < 0$), passing through $x = 0$ at $y = -3\sqrt{1} + 2 = -1$ and $x = 9$ at $y = -3\sqrt{4} + 2 = -4$. Domain $x \geq -3$.

Why the shift is 3 left, not 1. Written as $\frac{x}{3} + 1$ the constant is measured in the *stretched* input, so it must be divided by $b = \frac{1}{3}$ to be read in x : $1 \div \frac{1}{3} = 3$. Factoring does that division for you, which is why the factored form is the only one a shift can be read from directly.

Full credit requires the factored inside, the four parameters with input moves before output moves, a curve starting at $(-3, 2)$ and decreasing to the right, and the explanation of the 3-versus-1 shift. start $(-3, 2)$, falling, domain $x \geq -3$